# Zihe (Zach) Liu

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Distributed Reinforcement Learning (RL) for LLMs, Long-horizon Alignment

## EDUCATION

**University of Cambridge**
*Master of Engineering, Information and Computer Engineering* — *July 2026 - Present*
Masters Thesis: *Second Order Forward Optimization for Natural Policy Gradients*
- Develop efficient Gauss-Newton optimizer for RL, on custom OCaml ML compiler and AD stack (JAX-like)

*Bachelor of Arts, Information and Computer Engineering* — First Class Honors — *Sep 2023 - June 2026*
Coursework: RL, Deep Learning, Probabilistic Inference, Information Theory, Optimization, Computer Architecture
- David Newland Prize for Design (top undergraduate project)

## EXPERIENCE

**ERA: AI** — **Cambridge, UK**
*Research Fellow* — Mentor: Elizaveta Tennant, *Google DeepMind* — *July 2026 - Present*
- AI research fellowship (top 0.5% of 6,000+ applicants) exploring long-horizon RL alignment; 3 NeurIPS workshops
- Develop multi-turn RL algorithm to reduce ethical violations 17% at 2× training horizon (80K tokens, 128 turns)
- Scale distributed RL in verl (FSDP, async vLLM) on *4xB300*. Fused GatedDeltaNet kernels raised MFU 1.75%→27%
- Traced GPU hang to FlashInfer kernel deadlock, where fixing it enabled 4× higher batch-size, inference throughput

**Amazon** — **London, UK**
*Software Development Engineer Intern* — *July - Oct 2025*
- Built ML *(CatBoost)* triaging system to root-cause fatal Prime Video errors, with *AWS Bedrock* LLM log analysis
- Scaled throughput 100× (3→300 sessions/cluster, 20GB logs) by batching SQL queries on *AWS Lambda, Athena*
- Refactored monolithic *AWS CDK* stack, and unblocked a *Fargate* pipeline stalling deployment for 4 months
- *Turned down full-time return offer to work on RL research*

**UCL Centre for AI** — **London, UK**
*Student Researcher* — Mentor: Peter Hayes, *Humanloop (now Anthropic)* — *Feb - May 2025*
- Developed *Critique Direct Preference Optimization*: generative critique for sample-efficient DPO | Blog
- Distributed training on *Qwen-2.5* 3B models with *16x H100 GPUs*, on UCL HPC facility | GitHub

**Singapore AI Safety Institute, Infocomm Media Development Authority** — **Singapore**
*Research Engineer Intern* — *Dec 2024 - July 2025*
- Research industry evaluation metrics for LLM reliability and robustness | Report
- Benchmark hallucination detectors on *HaluBench* (simple CoT LLM-judge outperforms all) | GitHub

**Alignment Research Engineer Accelerator (ARENA)** — **London, UK**
*Research Engineer* — 1 of 2 undergrads in a mostly-PhD cohort of ∼20 — *Sep - Oct 2024*
- Built GPT-2 LLM from scratch, reverse-engineered its internal reasoning circuits, and fine-tuned via RLHF (PPO)

**Cambridge AI Safety Hub** — **Cambridge, UK**
*Operations Lead, Student Researcher* — *Oct 2023 - Present*
- Organized AI safety workshops, hackathons, research reading groups on interpretability, alignment | R1 Talk

**Changi Airport Group** — **Singapore**
*Machine Learning Engineer Intern* — *July - Aug 2023*
- Trained P2PNet (object detection CNN model) to localize and count luggage trolleys *(MAE decreased by 5x compared to previous systems)*. Quantized model with ONNX to deploy on edge (Nvidia Jetson Nano)

**Republic of Singapore Air Force** — **Singapore**
*Security Trooper, Corporal First Class* — *Jan 2021 - Jan 2023*
- Led Immediate Response Force team (2IC) to win APGC Eagles Challenge 2022. Best Airman Award.

## PUBLICATIONS

**Z. Liu**, D. Le, D. Williams-King, E. Tennant. Multi-Turn RL for Long-Horizon Moral Alignment. *NeurIPS 2026 AI4GOOD, BiAlign, SocialAgent Workshops*
J. Robinson, L. Franceschetti, M. Lovett, D. Le, K. Sen, N. Narayanan, **Z. Liu**, E. Tennant. Reasoning–Behavior Coherence Beyond the Rewarded Context. *NeurIPS 2026 AIWILD Workshop*
M. Saghir, N. Raghavendra, **Z. Liu**, E. Gunter. NGD converges to less degenerate solutions than SGD. *arXiv*

## AWARDS

**SG Digital AI Scholarship** | 500k$+ Full undergraduate funding, for exceptional Singaporeans pursuing AI overseas
**GB3 RISC-V CPU Competition 2026** | *1st of 8 teams.* Optimize tiny CPU with instruction cache, lookaheads | GitHub
**SF2 Image Compression 2026** | *1st of 13 teams.* Built JPEG compression scheme with DCT, intra prediction | Results
**Changi Airport Group Scholarship** | Offered full undergraduate funding for fast-track graduate programme

## SKILLS

**Programming Languages:** Python, C/C++, OCaml, Verilog/SystemVerilog, TypeScript/JavaScript, MATLAB, Java
**ML Systems:** verl, TRL, vLLM, PyTorch FSDP, DeepSpeed-ZeRO, NumPy, JAX
**AWS/Cloud:** CloudFormation, CDK, Fargate, Lambda, S3, Athena
**Non-ML:** HPC (Sun Grid Engine), FPGA/Arduino/Embedded Systems, Git, SQL